

Artificial Intelligence and Women's Empowerment: Advancing Safety, Agency and Justice in the Digital Era

Himadri Gogoi

MA in Education, Cotton University, Assam

ABSTRACT

The twenty-first century marks the convergence of two transformative forces: the global movement for women's empowerment and the rapid evolution of artificial intelligence (AI). While women's empowerment continues to be a multidimensional struggle encompassing safety, economic independence, social justice, and agency, AI emerges as a tool with the potential to accelerate this process. Artificial Intelligence (AI) is reshaping societies across the globe, influencing governance, education, healthcare, law enforcement, and everyday life. For women, particularly in contexts marked by persistent gender inequality, AI presents both new opportunities and complex risks. Drawing from Indian and global contexts, it situates AI as a transformative tool capable of fostering empowerment, while also recognizing its ethical, cultural, and socio-political challenges. Case studies from India—including Safetipin, NCRB's AI-based crime mapping, and AI-enabled women's self-help groups (SHGs)—demonstrate how digital technologies are influencing women's lives. Global examples such as UN Women's AI initiatives and gender-bias mitigation in machine learning expand the comparative dimension. When implemented thoughtfully, AI can become more than a tool; it can be a catalyst for sustainable empowerment. It enables women to exercise autonomy, ensure protection and claim justice in social, economic and legal spheres.

KEYWORDS: Women's empowerment, Artificial Intelligence, Safety, Agency, Justice, India, Digital transformation, Gender equality

INTRODUCTION

Every transformative technology in history has reshaped human destiny. The printing press unlocked literacy, electricity redrew rhythms of daily life, and the internet dissolved barriers of geography and time. Today, artificial intelligence (AI) emerges as the defining force of the twenty-first century—capable of amplifying opportunity, but equally capable of reinforcing inequality. For women, whose lives continue to oscillate between remarkable progress and persistent marginalization, AI is not merely a tool; it is a stage where the struggles for safety, agency, and justice are rehearsed in new and complex forms. Philosophers and reformers across traditions have long emphasized that empowerment is central to human freedom. Simone de Beauvoir described women as historically constructed as the “Other,” denied full subjecthood in shaping society. Martha Nussbaum's capabilities approach advanced the idea that empowerment lies in the substantive freedoms to live with dignity, not in the formal granting of rights. In the Indian context, Swami Vivekananda stressed that no society can rise without the upliftment of women, urging education and strength as the foundations of liberation. These perspectives converge in

today's digital age, where empowerment means more than access to technology—it requires the capacity to shape and govern the technologies that define social, economic, and political life. Globally, the imprint of AI on women's lives is evident. Algorithms trained on vast datasets have improved early detection of breast and cervical cancer, potentially saving millions. AI-powered agricultural advisory systems enable women farmers in countries such as India and Kenya to adapt to shifting climate conditions. Machine-learning models assist governments in mapping gender-based violence hotspots to support targeted interventions. Yet beneath this potential lie fractures. According to UN Women (2023), women are 21% less likely than men to own a smartphone and 37% less likely to possess basic digital skills. This digital gap means that while AI accelerates progress for some, it risks deepening exclusion for many others.

India captures these contradictions powerfully. In urban areas, AI-enabled safety initiatives such as Safetipin crowdsource data to identify unsafe zones, while AI-based CCTV analytics monitor public transport hubs. Health-tech startups deploy machine learning to improve maternal health outcomes. However, rural India reflects the stubborn digital gender divide. The National Family Health Survey reveals that only one in three rural women uses the internet, compared to more than half of men. This exclusion deprives millions of access to AI-driven educational tools, agricultural advisory service and entrepreneurial platforms that could transform livelihoods.

Thus, AI in India is both an engine of empowerment and a mirror of inequality. Safety, traditionally understood as freedom from physical harm, now extends into digital spaces. AI-driven predictive policing is being piloted in several Indian states to anticipate areas vulnerable to crimes against women. While such innovations may deter violence, they also raise questions of privacy and potential misuse of surveillance data. Simultaneously, AI has been weaponized through deepfakes, online harassment, and targeted disinformation campaigns against women.

Data from the National Crime Records Bureau show that cybercrimes against women have nearly doubled in the last five years, underscoring the urgent need for safeguards. Agency—the ability to act and decide independently—requires not only that women access AI tools but also that they contribute to their creation and governance. Yet women account for less than 20% of AI professionals worldwide, with India lagging even further behind. This absence of representation means that the systems shaping the digital future are often designed without women's perspectives, perpetuating blind spots and biases. Justice, the third pillar, involves correcting such inequities. AI recruitment systems, for instance, have been shown to undervalue women's career trajectories, particularly penalizing career breaks taken for caregiving. Without deliberate attention to fairness, AI risks entrenching the very hierarchies it promises to dismantle.

Thus, AI does not automatically yield empowerment. For some women, it offers independence, healthcare, and safer mobility; for others, it magnifies exclusion, surveillance, and vulnerability. India's experience demonstrates that the real challenge is not the spread of AI itself, but the conditions of its use—who has access, who participates in its governance, and whose interests it serves. If inclusivity and fairness are woven into AI systems, India could pioneer a model of gender-just digital transformation; if ignored, it risks reinforcing divides that will echo far into the future.

Empowerment in the age of AI, therefore, must be understood as a deliberate act of justice. It is not the passive consequence of technological progress, but the outcome of social vision, policy choices, and cultural values. For women, this means moving beyond being passive subjects of AI-driven change to becoming active shapers of a more equitable digital future. The conversation on safety, agency, and justice in the age of AI must begin here—where the possibilities of technology intersect with the timeless struggle for equality and dignity.

OBJECTIVE OF THE STUDY

1. To examine the role of artificial intelligence in enhancing women's safety, agency, and justice in the digital era.
2. To identify challenges such as bias, exclusion, and cyber risks that may limit women's empowerment through AI.
3. To evaluate the role of policy, governance, and women's participation in shaping inclusive AI systems.
4. To propose pathways for building gender-sensitive and equitable AI-driven futures.

NEED OF THE STUDY

1. Women remain disproportionately vulnerable to violence, bias, and exclusion; AI can either mitigate or amplify these risks.
2. India's rapid digital expansion necessitates urgent integration of gender-sensitive approaches in AI development.
3. The global debate on AI ethics remains incomplete without focused attention on women's empowerment.
4. Understanding Indian and international case studies can help design inclusive policies and practices for AI governance.

METHODOLOGY

This study adopts a secondary research methodology. Peer-reviewed journals, UN Women and World Economic Forum reports, Indian government policy papers, and contemporary case studies are synthesized to construct a comprehensive perspective.

AI AND WOMEN'S SAFETY

AI technologies have been leveraged to enhance women's safety through various applications. Predictive policing tools analyze data to identify patterns of criminal activity, enabling law enforcement agencies to allocate resources effectively. AI-powered personal safety applications provide real-time location tracking and emergency alerts, offering women increased autonomy and security in public spaces. In India, the 'Garuda Drishti' initiative utilizes AI-enabled facial recognition systems to identify and apprehend individuals involved in human trafficking. This program demonstrates the potential of AI to address gender-based violence by enabling timely interventions. However, concerns regarding privacy, data security, and the potential for misuse highlight the need for ethical considerations in the deployment of such technologies.

AI AND WOMEN'S AGENCY

AI has the capacity to enhance women's agency by providing access to education, employment, and healthcare. Online learning platforms powered by AI offer personalized education experiences, enabling women to acquire skills and knowledge irrespective of geographical constraints. AI-driven healthcare applications facilitate remote consultations and personalized treatment plans, improving women's access to healthcare services. In the employment sector, AI can mitigate gender biases in recruitment processes by analyzing job descriptions and candidate profiles to ensure equitable opportunities. However, the effectiveness of these systems depends on the quality and inclusivity of the data used to train them. Biased

data can perpetuate existing inequalities, underscoring the importance of diverse representation in AI development.

AI AND JUSTICE

Access to justice is a fundamental aspect of women's empowerment. AI can play a pivotal role in promoting justice by streamlining legal processes, providing legal information, and facilitating access to legal representation. AI-powered chatbots and virtual assistants can offer preliminary legal advice, guiding women through complex legal systems. However, the implementation of AI in legal contexts raises concerns about accountability, transparency, and fairness. The opacity of algorithmic decision-making processes can undermine trust in legal systems and potentially lead to discriminatory outcomes. Ensuring that AI systems are transparent, accountable, and subject to oversight is essential to uphold the principles of justice.

ETHICAL CHALLENGES AND GOVERNANCE

The integration of AI into societal structures presents several ethical challenges. Algorithmic bias, data privacy, and the digital divide are critical issues that need to be addressed to ensure that AI serves as a tool for empowerment.

The lack of diversity in AI development teams can lead to the creation of systems that do not adequately consider the needs and perspectives of women. Governance frameworks for AI must be inclusive, transparent, and accountable. Policies should be developed to regulate the use of AI, ensuring that it is deployed ethically and equitably. International collaboration is necessary to establish standards and guidelines that promote the responsible use of AI technologies.

CASE STUDIES

GLOBAL INITIATIVES

1. **AI-Powered Personal Safety Applications:** Applications such as 'BSafe' and 'SafeTrek' utilize AI to provide real-time location tracking and emergency alerts, enhancing women's safety in public spaces.
2. **Predictive Policing Tools:** In the United States, predictive policing tools analyze crime data to identify patterns and allocate resources effectively, aiming to prevent gender-based violence.
3. **AI in Healthcare:** AI-driven healthcare applications, such as 'Ada' and 'Babylon Health,' offer personalized medical advice and remote consultations, improving women's access to healthcare services.

INDIAN INITIATIVES

1. **Garuda Drishti:** The Maharashtra Police's 'Garuda Drishti' initiative employs AI-enabled facial recognition systems to identify and apprehend individuals involved in human trafficking, demonstrating the potential of AI to combat gender-based violence.
2. **AI in Education:** Platforms like 'BYJU'S' and 'Vedantu' offer AI-powered personalized learning experiences, enabling women, particularly in rural areas, to access quality education.
3. **AI in Healthcare:** The 'eSanjeevani' telemedicine platform utilizes AI to provide remote consultations, improving healthcare access for women in underserved regions.

FINDINGS

1. AI can significantly enhance women's empowerment but risks reinforcing inequalities if not inclusively designed.
2. Indian innovations like Safetipin demonstrate that context-sensitive AI tools can improve safety and urban planning.
3. Women's participation in AI development remains limited, reducing opportunities to embed gender-sensitive perspectives.
4. Governance frameworks are evolving but often reactive rather than anticipatory.
5. Empowerment requires integrating AI with broader social reforms in education, healthcare, and policy.

CONCLUSION

Artificial intelligence is neither inherently liberating nor oppressive; it is a tool whose outcomes depend on human choices. For women's empowerment, the promise of AI lies in its ability to expand safety, agency, and justice. Yet, this promise is fragile, threatened by bias, exclusion, and misuse. India, with its complex socio-cultural landscape, has both unique challenges and opportunities: challenges in bridging digital divides, and opportunities in designing inclusive, context-driven AI solutions. The active participation of women in AI development is essential to create systems that genuinely serve their needs and promote gender equality. As we navigate the complexities of the digital era, it is imperative to harness AI as a tool for positive change, ensuring that it contributes to the realization of women's rights and empowerment. The future of women's empowerment in the digital age must be rooted in inclusive design, ethical governance, and women's participation as creators—not just users—of technology. In reimagining empowerment, one must heed Sarojini Naidu's vision: "Awake, arise, and assert yourselves, be not afraid." AI, when guided by this spirit, can help women assert themselves in every sphere of life—safely, justly, and with agency.

REFERENCES

1. Nussbaum, M. (2000). *Women and human development: The capabilities approach*. Cambridge University Press.
2. Sen, A. (1999). *Development as freedom*. Oxford University Press.
3. Buolamwini, J. (2023). *Unmasking AI: The impact of algorithmic bias on marginalized communities*. Algorithmic Justice League.
4. Noble, S. U. (2018). *Algorithms of oppression: How search engines reinforce racism*. NYU Press.
5. Shah, S. S. (2024). Gender bias in artificial intelligence: Empowering women through digital literacy. *Premier Journal of Artificial Intelligence Review*. Retrieved from <https://premierscience.com>.
6. Richardson, P. E. (2024). Evaluating a women's digital inclusion and storytelling project in Zimbabwe. *ScienceDirect*. Retrieved from <https://www.sciencedirect.com>
7. NITI Aayog. (2018). *National Strategy for Artificial Intelligence*. Government of India.
8. Nussbaum, M. (2000). *Women and Human Development: The Capabilities Approach*. Cambridge University Press.
9. Safetipin (2021). *Annual Impact Report*. Safetipin.org.
10. UN Women. (2020). *Artificial Intelligence and Gender Equality*. UN Women Publications.
11. World Bank. (2021). *Reimagining Digital Inclusion in South Asia*. Washington, DC.
12. Sinha, A. (2022). AI for Social Good in India. *Economic & Political Weekly*, 57(32), 15–20.

13. Choudhury, N. (2021). Gender Bias in AI Systems. Indian Journal of Gender Studies, 28(2), 150–172.
14. <https://www.google.com>